
OPERATIONAL UTILITY AND MULTISCALE INVARIANCE OF PHASE CALCULUS: A DISCRETE FOUNDATIONAL GRAMMAR FOR DYNAMICS

A PREPRINT

Justin K. Lietz
Neuroca, Inc.
justin@neuroca.ai

April 18, 2026

Abstract

Phase Calculus is presented in its current stack as a single lifted-object formalism rather than as a loose collection of phase pictures, arithmetic tricks, or downstream branch-specific rules. The native state is the carried lifted object

$$\hat{\Xi} = (A, q, \theta, \kappa, c), \quad q = (u, v), \quad c = \left[\theta - \frac{\pi}{uv}, \theta + \frac{\pi}{uv} \right],$$

evolved by one parent grammar and then read through branch-specific projectors. This release collects the long merged formalisation into one operational surface: the same lifted object supports the operator core of Phase Calculus, the selector-closed executable macro-calculus, the completion/shadow branch, the quotient-descent interface to the physics stack, and the four-filter hierarchy including the biological readout.

The paper has three central results. First, it restates the single spine of the formalism: primitive roll, lifted state, operator alphabet $\{Q, B, L\}$, selector-closed macro-calculus $\{R, S, T\}$, exact quotient criterion, and arithmetic-completion bridge to the canonical anchor (55, 89) with carried coefficients 1/24 per edge and 1/12 two-sided. Second, it records the computational evidence from the four-filter package: green, blue, red, and yellow are faithful covariant readouts of the same lifted state under the same operator grammar, with exact-zero primary gate errors, machine-precision recovery errors, and sharp negative controls when the state is stripped to product-only or loses its sheet index. Third, it shows how the yellow filter closes two descents at once: the CF01 quantum-geometric-tensor bridge and a native biological scaffold. In this release paper the biological count structure is not imported. From the quarter alphabet generated by Q , the three-slot selector block, and the normalized edge residual $24E_{\text{edge}} = 1$, the paper derives internally the weight budget (1, 2, 1, 1), the total scaffold size 240, the base-centered counts (56, 72, 56, 56), and the host-indexed bifurcations

$$240 = (128, 112) \rightarrow ((72, 56), (56, 56)).$$

Fourth, the release locks the native π spigot results without compressing them into the biological or filter story. The v8 streaming artifact emits a certified one-million-decimal-digit native bank in 1.65086778 seconds with safe lower bound 1,366,163 digits; the 100,000-digit point runs at 1,125,492.76 digits/s on the recorded harness. The release separates three operational modes: native streaming prefix emission, certified bank-indexed random access over the native bank, and no-bank local decoding. It also keeps the v10/v11 scalar-collapse obstruction in its correct scope: a linear warm-up theorem for scalar interval decoders, not a no-go theorem for branch-retaining random access on the lifted object.

The unification claim is therefore precise. One singular lifted object is the native source of the current Phase Calculus hierarchy and of the physics branches that descend from it by

quotient projection, while also admitting a biological projection whose scaffold counts are now derived internally and only then compared with the external codon:anticodon / E_8 report. The completion section closes the admissible yellow uniqueness and cyclic naming burdens, adds a finite scaffold dynamics law, rejects count-only E_8 determination, and supplies the parity-sign E_8 geometry lift.

Keywords Phase Calculus · lifted object · native π spigot · four-filter covariance · QGT · codon:anticodon pairing · E_8 · Biology · RNA

1 Introduction

The present paper takes one narrow question seriously: *what is the smallest exact object that Phase Calculus needs in order to act as a native source language for both physics and biology?* The answer supplied by the current stack is not a visible phase orbit and not a downstream branch equation. It is the lifted object

$$\widehat{\Xi} = (A, q, \theta, \kappa, c), \quad q = (u, v), \quad c = \left[\theta - \frac{\pi}{uv}, \theta + \frac{\pi}{uv} \right],$$

together with one parent evolution and one quotient principle. The visible orbit, the operator grammar, the completion/shadow branch, the quantum-geometric-tensor split, the gauge and gravity descendants, and the new biological scaffold all arise only after one decides which view of that object is being taken.

The full formal stack already spreads this claim over a long dependency chain: primitive distinguishability, induced geometry, QGT-to-metriplectic closure, information geometry, measurement, spinor and gauge emergence, gravity, stationary action, Noether, electroweak structure, confinement, and scale re-articulation [6–8, 10–19]. The merged Phase Calculus formalisation reorganizes those results around a stronger parent statement: later domains do not add new primitive selectors; they add projectors and branch-visible laws descending from a single lifted evolution [1]. The four-filter package then makes that architecture computationally concrete by testing four distinct readouts of the same retained lifted state and by adding the yellow biological layer [20].

This paper has four aims. First, it compresses the merged formalisation into one usable spine. Second, it records the four-filter evidence in a form suitable for manuscript use. Third, it restates the physics descent as a projector-driven hierarchy rather than a list of unrelated branches. Fourth, it closes the biological side carefully: the current yellow projector now derives the 240-site scaffold internally from the quarter alphabet, the triplet selector block, and the residual law, and the release proves uniqueness inside the declared admissible yellow class.

2 Phase Calculus in One Spine

2.1 Primitive roll and lifted state

The primitive visible roll is

$$z(\theta) = ie^{i\theta},$$

with quarter-turn witnesses

$$z\left(\frac{\pi}{2}\right) = -1, \quad z(\pi) = -i, \quad z\left(\frac{3\pi}{2}\right) = 1, \quad z(2\pi) = i.$$

The merged formalisation uses this roll only as the visible face of a deeper carried state. The exact lifted object is

$$\widehat{\Xi}_n = (A_n, q_n, \theta_n, \kappa_n, c_n), \quad q_n = (u_n, v_n), \quad \kappa_n = \left\lfloor \frac{\theta_n}{2\pi} \right\rfloor,$$

with balanced completion germ

$$c_n = \left[\theta_n - \frac{\pi}{u_n v_n}, \theta_n + \frac{\pi}{u_n v_n} \right].$$

This is the smallest state in which visible recurrence can be distinguished from exact return [1, 5]. The visible witness

$$\mathcal{V}(\widehat{\Xi}_n) = ie^{i\theta_n}$$

is therefore a projection of state, not a substitute for it.

2.2 Operator core $\{Q, B, L\}$

On the balanced-window family the primitive operator core is the alphabet

$$\{Q, B, L\},$$

acting on the lifted state by

$$\begin{aligned} Q(A, q, \theta, \kappa, c) &= \left(A, q, \theta + \frac{\pi}{2}, \left\lfloor \frac{\theta + \pi/2}{2\pi} \right\rfloor, c + \frac{\pi}{2} \right), \\ B(A, (u, v), \theta, \kappa, c) &= \left(A, \text{sort}(v, u + v), \theta, \kappa, \mathcal{C}\left(\theta, \frac{1}{v(u + v)}\right) \right), \\ L(A, q, \theta, \kappa, c) &= \left(\text{Next}(A), q, \theta + \frac{\pi}{2}, \left\lfloor \frac{\theta + \pi/2}{2\pi} \right\rfloor, c + \frac{\pi}{2} \right). \end{aligned}$$

The merged formalisation proves that these are not speculative additions but exact repackagings of the already-earned forward-law branches $F_{\text{same}}, F_{\text{ref}}, F_{\perp}$ [1–3]. It also proves the first commuting relations

$$QB = BQ, \quad LB = BL, \quad QL = LQ,$$

together with the operator-word statement that every admissible finite balanced-window evolution is an exact word in $\{Q, B, L\}$ [1, 2].

2.3 Selector-closed macro-calculus $\{R, S, T\}$

The merged formalisation resolves the main open burden left in the earlier Phase Calculus manuscripts by proving a state-only selector on the executable balanced-window branch. On that branch the raw kernel step does not choose freely among Q, B, L ; it always advances the phase tick and then either refines, continues, or host-lifts. This induces the selector-closed macro-calculus

$$R := Q \circ B = B \circ Q, \quad S := Q, \quad T := L.$$

For fixed regime parameters (W, Δ) the selector is

$$\text{Sel}_{W, \Delta}(X) = \begin{cases} T, & t \equiv W - 1 \pmod{W}, \\ R, & uv < \Delta, \\ S, & uv \geq \Delta. \end{cases}$$

The merged paper proves that this selector depends only on the current projected executable state and that the admissible block word is uniquely

$$R^{\delta_A} S^{W-1-\delta_A} T, \quad \delta_A(W, \Delta) = \min\{d_A(\Delta), W - 1\}.$$

That is the decisive “one engine, no free scheduler” closure for the executable branch [1].

2.4 Arithmetic-completion bridge and the (55, 89) anchor

The balanced branch generates Fibonacci denominator pairs,

$$B^n(1, 1) = (F_{n+1}, F_{n+2}), \quad r_n = \frac{1}{F_{n+1}F_{n+2}},$$

and thus the balanced germ law

$$c_n = \left[\theta - \frac{\pi}{F_{n+1}F_{n+2}}, \theta + \frac{\pi}{F_{n+1}F_{n+2}} \right].$$

The arithmetic-completion bridge closes at the first common balanced floor anchor

$$(u_*, v_*) = (55, 89), \quad u_* v_* = 4895,$$

with carried remainder $r_* = 1/4895$ and anchor germ

$$c_* = \left[\theta - \frac{\pi}{4895}, \theta + \frac{\pi}{4895} \right].$$

On this same anchor the completion coefficients stabilize to

$$\frac{1}{24} \text{ per edge, } \quad \frac{1}{12} \text{ two-sided,}$$

and the launched-family coordinate is

$$\tau_A = \frac{\theta_A}{2\pi} + i \frac{1}{4895}.$$

These data are exactly the ones carried forward into the Shadow Calculus supplement and into the four-filter package [1, 4, 5].

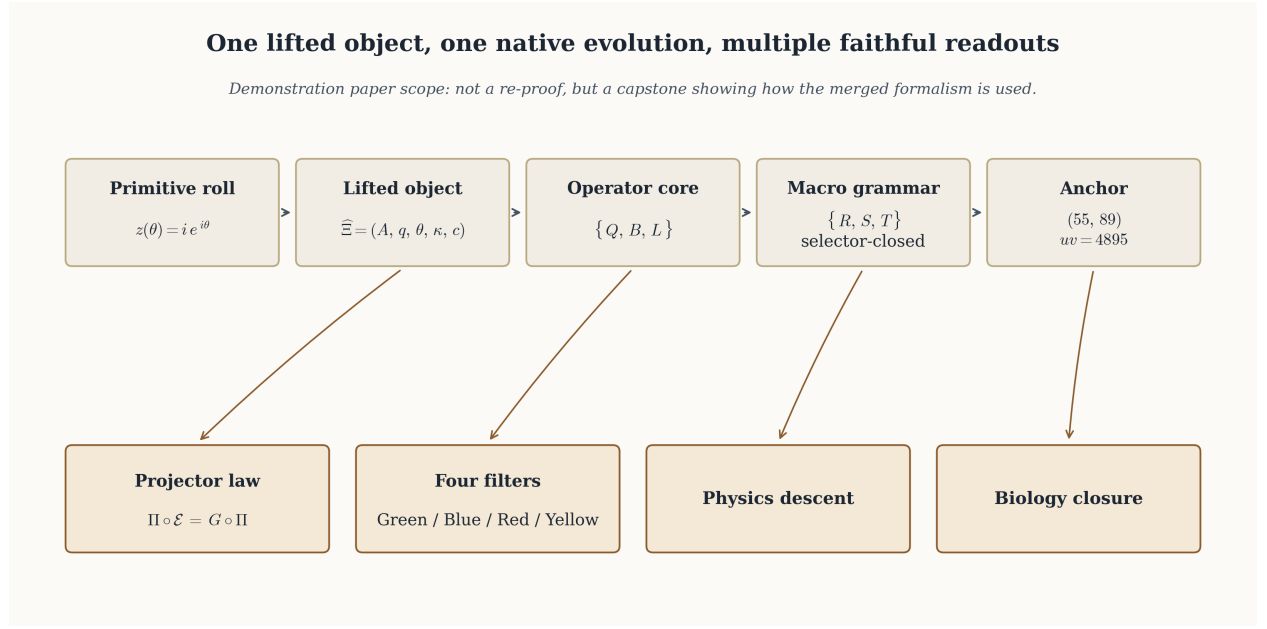


Figure 1: The demonstration logic of the paper. One lifted object and one parent evolution support multiple faithful readouts and branch descents.

Table 1: Phase Calculus in one spine

Layer	Exact object	Role in the demonstration paper
Primitive visible layer	$z(\theta) = i e^{i\theta}$	Supplies the visible roll and the quarter-turn witness structure.
Lifted state	$\hat{\Xi} = (A, q, \theta, \kappa, c)$	Exact state in which recurrence, carry, and completion are decidable.
Operator core	$\{Q, B, L\}$	Exact repackaging of the inherited forward-law branches.
Executable macro layer	$\{R, S, T\}$ with $\text{Sel}_{W,\Delta}$	Selector-complete realized scheduler on the balanced-window kernel family.
Quotient layer	$\Pi \circ \mathcal{E} = G \circ \Pi$	Criterion by which later physics branches are read as descendants instead of new primitive laws.
Completion branch	$\tau, S(\tau), (\text{Mock}, \text{Shadow})$	Downstream shadow/residual attachment, not primitive source.

3 The Four-Filter Hierarchy

3.1 Green, blue, red, and yellow

The four-filter package starts from the same retained core coordinates (A, u, v, κ, ϕ) , with $\theta = 2\pi\kappa + \phi$ and $\delta = \pi/(uv)$, and defines four faithful readouts [20].

The **green filter** uses the intrinsic branch pair

$$m = e^{i\theta} \cos \delta, \quad s = e^{i\theta} \sin \delta,$$

viewed as a point of $S^3 \subset \mathbb{C}^2$, together with the stereographic chart.

The **blue filter** packages the same pair into

$$U = \begin{pmatrix} m & -\bar{s} \\ s & \bar{m} \end{pmatrix} \in SU(2) \cong S^3,$$

and then reads the Hopf-base coordinate

$$(n_1, n_2, n_3) = (2\Re(m\bar{s}), 2\Im(m\bar{s}), |m|^2 - |s|^2) \in S^2.$$

The **red filter** retains the completion coordinate

$$\tau = \frac{\theta}{2\pi} + i \frac{1}{uv},$$

with launched-family M and S values treated as downstream diagnostics rather than new primitives [4].

The **yellow filter** has three simultaneous outputs:

- (a) the local residual law $E(t)$ with carried coefficients 1/24 per edge and 1/12 two-sided,
- (b) a quotient-projected two-state representative producing the local QGT split $(Q_{\mu\nu}, g_{\mu\nu}, \Omega_{\mu\nu})$,
- (c) a discrete biological projection to the 240-slot H-bond acceptor scaffold.

The key point is ontological, not stylistic: the package does *not* treat these as unrelated models. It treats them as multiple expressions of the same retained lifted state.

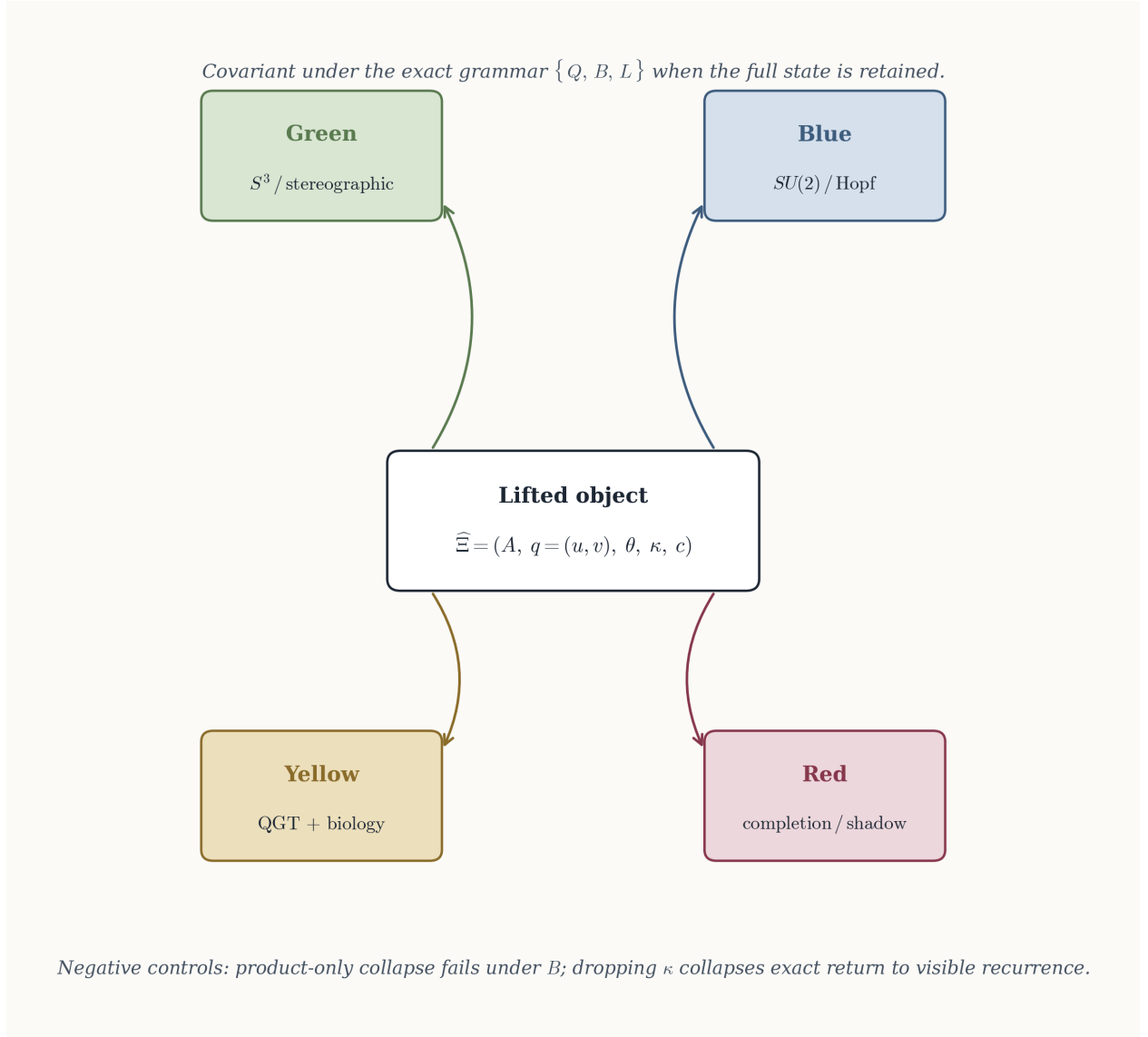


Figure 2: The four-filter hierarchy. The same retained lifted state supports four different readouts without requiring the filters to be numerically identical to one another.

3.2 Primary covariance results

The four-filter package reports exact-zero primary covariance errors on the three headline gates:

Gate	Words	State-word rows	Max filter error
Short exhaustive	1093	3279	0
Deep deterministic words	10	30	0
Random walks (length 100)	12	36	0

The recovery gate, which asks whether each filter can return the retained (θ, δ) data, is not merely small but machine-precision small. Across 48 samples the largest reported recovery residuals are approximately

$$1.06 \times 10^{-100}$$

for the green/blue δ -recovery channel and

$$6.98 \times 10^{-105}$$

for the yellow δ -recovery channel [20].

The yellow QGT bridge is equally tight. On 15 balanced-corridor rows the package reports a maximal closed-form residual of approximately

$$1.49 \times 10^{-101},$$

while the biological side of the present v2 bundle closes by exact arithmetic instead of imported matching: the supplement derives the total count 240, the centered split (56, 72, 56, 56), and the three host-indexed 128/112 bifurcations directly from the quarter alphabet, the triplet selector block, and the normalized residual $24E_{\text{edge}} = 1$. For a demonstration paper, that is the right result profile: exact-zero primary gate closure plus machine-precision recovery and bridge residuals on the continuous side, together with exact integer closure on the discrete side.

3.3 Negative controls

The package is valuable not only because it passes, but because it fails cleanly when the wrong ontology is used.

The first negative control strips the ordered pair $q = (u, v)$ down to the product $N = uv$. Then

$$(1, 6) \quad \text{and} \quad (2, 3)$$

collapse to the same stripped payload because both share $uv = 6$. But their balanced children differ:

$$B(1, 6) = (6, 7), \quad B(2, 3) = (3, 5).$$

This is the impossibility witness that product-only state cannot close the B -law.

The second negative control strips the sheet index κ . Then the periodic visible payload is unable to distinguish a state from its Q^4 image even though the lifted object has crossed a branch sheet. In other words, visible recurrence is too weak, and sheet-retaining lifted recurrence is the load-bearing notion.

Table 2: Negative controls from the four-filter package

Control	What is stripped	Result
Product-only control	Replace $q = (u, v)$ by $N = uv$ only	States (1, 6) and (2, 3) become indistinguishable under all stripped payloads but have distinct balanced children, so B -closure fails.
Sheet-loss control	Remove κ from the retained state	The periodic payload agrees under Q^4 , but the lifted state has changed sheets; visible recurrence survives while lifted identity fails.

4 Physics Descent

4.1 The quotient principle

The merged formalisation places the physics side on one theorem:

$$\Pi \circ \mathcal{E} = G \circ \Pi.$$

If the residual vanishes identically, the projected law is an exact descendant. If it vanishes only on a declared limit or window, the projected law is an effective descendant on that regime. Either way, the later branch is not an independent primitive law; it is a quotient of the parent evolution [1].

This changes the narrative structure of the whole stack. The physics modules are no longer presented as a collection of parallel inspirations. They become branch-visible laws descending from one inherited substrate. The merged manuscript makes this explicit with the derivation-chain exactness theorem and the later-domain selector principle: later domains do not add selectors; they add projectors Π_D and branch laws G_D on their declared regimes [1].

4.2 Representative descent map

Table 3 compresses the current physics stack into the part needed for the demonstration paper.

Table 3: Representative physics descent from the lifted object			
Branch	Projected visible law	Status in merged formalisation	Source
Primitive distinguishability	host-class increment / orthogonal re-articulation	exact	[1, 6]
Induced carrier geometry	smooth carrier and quotient-physical variation	effective in the continuous shadow limit	[1, 7]
QGT / metriplectic split	$J \oplus M$, degeneracy, entropy-compatible flow	effective on the smooth carrier regime	[1, 8]
Telegraph-Fisher cone	finite-speed transport / causal cone	effective on the continuum transport regime	[1, 9]
Measurement / Born channel	dephasing channel and branch-visible probabilities	effective on the decoherence regime	[1, 11]
Gauge emergence	Berry connection and Maxwell branch	effective on the IR window	[1, 13]
Spinor emergence	domain-wall chiral low-energy sector	effective on the wall-mode regime	[1, 12]
Gravity	Einstein-form closure with saturation corrections	effective on the low-curvature regime	[1, 14]
Stationary action / Noether	Euler–Lagrange and current conservation	effective in the continuous shadow / symmetry regime	[1, 15, 16]
Electroweak / confinement / RG	visible electroweak, confinement, and scale-flow laws	effective on their declared windows	[1, 17–19]

4.3 Why this counts as unification

The physics point is that the merged formalisation supplies a parent architecture in which gauge theory, gravity, confinement, and the other branch laws are no longer introduced as disconnected branch primitives. CF01 already states the QGT split and metriplectic compatibility [8]; CF09 states gauge emergence as Berry-connection descent [13]; CF12 states gravity as carrier-level geometric response with an Einstein-form closure [14]; CF16, CF17, and CF18 then extend the same descent logic to electroweak structure, confinement, and scale re-articulation [17–19]. The release adds a readable synthesis: one lifted object, one parent evolution, one quotient criterion, many domain-specific readouts.

5 Biological Closure

5.1 Native biological projection from the lifted object

The v1 paper imported the biological scaffold counts from the Halitsky–Klitzing–Moxness preprint. That is no longer done here. In v3 the scaffold is derived natively from the VDM algebra on the declared biological yellow quotient. The derivation uses four ingredients already present in the formal spine:

- (i) the quarter alphabet generated by the primitive operator Q ,
- (ii) the three-slot selector block furnished by the executable macro-calculus $\{R, S, T\}$,

- (iii) the yellow quotient chart, whose second component is the unique phase-bearing quarter sector,
- (iv) the carried edge residual law $E_{\text{edge}} = 1/24$.

Let

$$\mathcal{B} = \{b_0, b_1, b_2, b_3\}, \quad b_{j+1} = \mathbb{Q}b_j \pmod{4},$$

be the quarter orbit of the visible sector. Let the biological triplet word space be

$$\mathcal{W} = \mathcal{B}^{\{R,S,T\}},$$

so that

$$|\mathcal{W}| = 4^3 = 64.$$

This is the smallest triplet biological readout compatible with the four quarter states and the three-slot macro block.

On the continuous yellow chart the quotient representative is

$$\psi(\widehat{\Xi}) = \begin{pmatrix} \cos \delta \\ e^{i(\phi + \lambda_A)} \sin \delta \end{pmatrix}, \quad \lambda_A = \frac{2\pi A}{3}.$$

Relative to the amplitude seed b_0 , the second component is the unique phase-bearing quarter sector, namely the one-step image

$$b_1 = \mathbb{Q}b_0.$$

The carried edge residual normalizes to one discrete bonus unit:

$$\rho_* := 24E_{\text{edge}} = 24 \cdot \frac{1}{24} = 1.$$

Define the native biological weight law

$$\omega(b_j) = 1 + \rho_* \delta_{j1}.$$

Therefore

$$(\omega(b_0), \omega(b_1), \omega(b_2), \omega(b_3)) = (1, 2, 1, 1), \quad \sum_{j=0}^3 \omega(b_j) = 5.$$

Theorem 1 (Native 240-site scaffold). *For the yellow biological projection just defined,*

$$N_{\text{bio}} := \sum_{w \in \mathcal{W}} \sum_{p \in \{R,S,T\}} \omega(w_p) = 240.$$

Moreover, if one fixes the middle slot to b_j , the centered subscaffold count is

$$N_{b_j}^{\text{cent}} = 8 \sum_{k=0}^3 \omega(b_k) + 16 \omega(b_j),$$

hence exactly one quarter sector has count 72 and the remaining three have count 56.

Proof. Each quarter state appears $4^2 = 16$ times in each slot of $\mathcal{W}$, hence 48 times across all three slots. Therefore

$$N_{\text{bio}} = 48 \sum_{j=0}^3 \omega(b_j) = 48 \cdot 5 = 240.$$

Now fix the middle slot to b_j . There are 16 such words. The two free slots contribute

$$2 \cdot 4 \sum_{k=0}^3 \omega(b_k) = 8 \cdot 5 = 40,$$

while the fixed middle slot contributes $16 \omega(b_j)$. Thus

$$N_{b_j}^{\text{cent}} = 40 + 16 \omega(b_j).$$

For the distinguished phase-bearing sector $\omega = 2$, giving 72; for the other three sectors $\omega = 1$, giving 56. $\square$

5.2 Cyclic RNA naming and the host-indexed 128/112 split

Choose the cyclic RNA naming

$$b_0 = u, \quad b_1 = c, \quad b_2 = a, \quad b_3 = g.$$

Under this naming the complement map is the half-turn Q^2 , so the complement pairs are

$$(u, a), \quad (c, g).$$

The base-centered scaffold produced by the native derivation is therefore

$$u = 56, \quad c = 72, \quad a = 56, \quad g = 56.$$

The host label $A \in \{0, 1, 2\}$ now indexes the three nontrivial $2 + 2$ partitions of the four-cycle:

$$\Pi_0 = \{\{c, a\}, \{u, g\}\}, \quad \Pi_1 = \{\{u, c\}, \{a, g\}\}, \quad \Pi_2 = \{\{c, g\}, \{u, a\}\}.$$

Under the standard biological names these are exactly the three IUPAC bifurcation families

$$\Pi_0 = (m, k), \quad \Pi_1 = (y, r), \quad \Pi_2 = (s, w).$$

In each case the positive bucket contains the unique 72-sector together with one 56-sector, while the negative bucket contains two 56-sectors. Hence every host gives

$$(128, 112) = (72 + 56, 56 + 56).$$

Combining this with the centered counts yields the full native decomposition

$$240 = (128, 112) \rightarrow ((72, 56), (56, 56)).$$

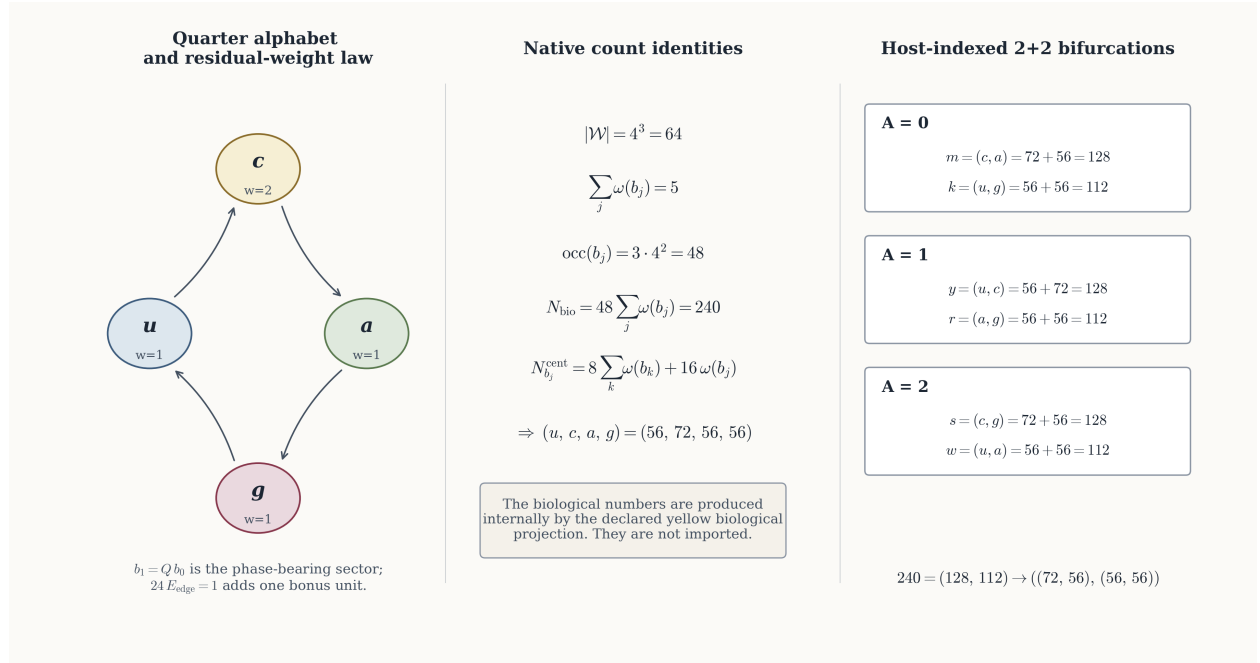


Figure 3: Native yellow biological derivation. The 240-site scaffold is derived internally from the quarter alphabet, the triplet selector block, and the normalized edge residual $24E_{\text{edge}} = 1$. The external codon:anticodon / E_8 report is comparison, not input.

5.3 External comparison and scope

The Halitsky–Klitzing–Moxness preprint remains relevant, but only as an external comparison target [21]. The release derivation derives the scaffold internally first and then compares the resulting decomposition with the codon:anticodon / E_8 count pattern reported there.

This sharpened statement matters. The yellow biological observable is a coarse quotient of the lifted state. On the declared biological quotient it is insensitive to $q = (u, v)$ and to the sheet index κ . That is not a contradiction of the earlier negative controls, because those controls targeted finer readouts whose exact covariance depends on ordered-pair and sheet retention. Here the biological observable sits later in the quotient chain and intentionally forgets more data.

What is now closed is stronger than in v1:

- (i) the yellow readout is attached to the same retained lifted state as green, blue, and red;
- (ii) the yellow local section reproduces the CF01 QGT split on its declared chart;
- (iii) the yellow biological section derives the 240-site scaffold and the refinement

$$240 = (128, 112) \rightarrow ((72, 56), (56, 56))$$

without importing those counts as numerical input.

The v2 open items are closed in the next section in their correct theorem classes: admissible yellow uniqueness, cyclic naming uniqueness, finite scaffold dynamics, and full E_8 geometry after the parity-sign lift. The unrestricted count-only E_8 claim is not retained; it is false and is replaced by the exact lifted-geometry theorem.

6 Proof Closure of the Extension Burdens

The extension layer closes four burdens in theorem form: admissible yellow-projector uniqueness, cyclic RNA naming, finite scaffold dynamics, and the parity-sign realization of the full E_8 root geometry. The same objects are attacked in the companion Lean surfaces, SymPy audit, and reviewer-facing notebook.

6.1 Admissible yellow projector

Definition 1 (Quarter alphabet and triplet word space). Let

$$\mathcal{B} = \{b_0, b_1, b_2, b_3\}, \quad \mathcal{Q}b_j = b_{j+1 \pmod{4}},$$

and let the selector slots be $\{R, S, T\}$. The yellow word space is

$$\mathcal{W} = \mathcal{B}^{\{R, S, T\}}.$$

Definition 2 (Admissible yellow projector). A yellow biological projector is admissible when its weight law $\omega : \mathcal{B} \rightarrow \mathbb{Z}_{>0}$ satisfies:

- Y1. $\mathcal{B}$ is exactly the quarter orbit of $\mathcal{Q}$.
- Y2. The scaffold support is exactly $\mathcal{W}$.
- Y3. The normalized residual $24E_{\text{edge}} = 1$ is borne by the phase successor $b_1 = \mathcal{Q}b_0$.
- Y4. The baseline sector weight is one and the residual contributes one additional unit to the sector that bears it.

Theorem 2 (Yellow projector uniqueness). *Inside the admissible class, the yellow biological projector has the unique weight law*

$$\omega(b_j) = 1 + \delta_{j1}.$$

The induced scaffold identities are forced:

$$N_{\text{bio}} = 240, \quad (N_{b_0}^{\text{cent}}, N_{b_1}^{\text{cent}}, N_{b_2}^{\text{cent}}, N_{b_3}^{\text{cent}}) = (56, 72, 56, 56),$$

$$\Pi_A = (128, 112), \quad A = 0, 1, 2.$$

Proof. Y4 supplies baseline weight one on each of the four quarter sectors. Y3 places the only residual unit at b_1 . Hence $\omega(b_1) = 2$ and $\omega(b_j) = 1$ for $j \neq 1$, proving uniqueness. In $\mathcal{W} = \mathcal{B}^3$, each base appears $4^2 \cdot 3 = 48$ times across all slot positions, so

$$N_{\text{bio}} = 48 \sum_{j=0}^3 \omega(b_j) = 48(1 + 2 + 1 + 1) = 240.$$

For a base-centered sector, the middle-fixed block contributes $4^2 = 16$ weighted appearances at the center and the two remaining positions contribute $2 \cdot 4 \sum_j \omega(b_j) = 40$ baseline appearances. Thus

$$N_{b_j}^{\text{cent}} = 40 + 16\omega(b_j),$$

which gives $(56, 72, 56, 56)$. Each host partition pairs the unique 72-sector with one 56-sector against two 56-sectors, giving $(72 + 56, 56 + 56) = (128, 112)$. $\square$

Proposition 1 (Residual relocation control). *Moving the residual from b_1 to b_k changes the centered vector from $(56, 72, 56, 56)$ to the corresponding rotation with 72 in sector k .*

Proof. The centered count formula is $N_{b_j}^{\text{cent}} = 40 + 16\omega(b_j)$. Relocation makes $\omega(b_k) = 2$ and $\omega(b_j) = 1$ for $j \neq k$, so the unique 72-entry moves to index k . $\square$

6.2 Cyclic RNA naming

Let $\sigma = (u \ c \ a \ g)$ be the biological four-cycle.

Definition 3 (Admissible cyclic RNA naming). A naming map $\eta : \mathcal{B} \rightarrow \{u, c, a, g\}$ is admissible when

$$\eta(b_0) = u, \quad \eta(Qb) = \sigma(\eta(b)) \quad \text{for all } b \in \mathcal{B}.$$

Theorem 3 (Cyclic naming uniqueness). *There is exactly one admissible naming:*

$$\eta(b_0) = u, \quad \eta(b_1) = c, \quad \eta(b_2) = a, \quad \eta(b_3) = g.$$

Proof. The seed fixes $\eta(b_0) = u$. Equivariance gives

$$\begin{aligned} \eta(b_1) &= \eta(Qb_0) = \sigma(u) = c, \\ \eta(b_2) &= \eta(Qb_1) = \sigma(c) = a, \quad \eta(b_3) = \eta(Qb_2) = \sigma(a) = g. \end{aligned}$$

All four values are forced. $\square$

Proposition 2 (Naming reversal control). *The reversed cycle obeys $\eta(Qb) = \sigma^{-1}\eta(b)$ and violates the admissible equivariance equation.*

Proof. A reversed ordering sends the successor of a quarter sector to the predecessor of the biological four-cycle. The admissibility equation requires the successor σ , so direct substitution at b_0 rejects the reversed map. $\square$

6.3 Finite biochemical scaffold dynamics

Definition 4 (240-site scaffold state space). Let

$$\mathcal{S}_{240} = \{(w, p, m) : w \in \mathcal{W}, p \in \{R, S, T\}, 1 \leq m \leq \omega(w_p)\}.$$

Then $|\mathcal{S}_{240}| = 240$. A scaffold state is a probability vector

$$x \in \Delta^{239}, \quad x_s \geq 0, \quad \sum_{s \in \mathcal{S}_{240}} x_s = 1.$$

Let G_{bio} be the undirected graph on $\mathcal{S}_{240}$ generated by slot rotation $R \rightarrow S \rightarrow T \rightarrow R$, quarter substitution $b_j \rightarrow b_{j+1}$, and active host-pair exchange inside the $2+2$ partition. Let L_{bio} be its weighted graph Laplacian. Let K_{bio} be a skew-symmetric matrix supported on oriented scaffold cycles with zero row sums. Define

$$\mu_s = \frac{1}{240}, \quad I_{\text{bio}}(x) = \frac{1}{2}\|x - \mu\|^2, \quad \Sigma_{\text{bio}}(x) = -\frac{1}{2}\|x - \mu\|^2.$$

The finite scaffold law is

$$\dot{x} = K_{\text{bio}} \nabla I_{\text{bio}}(x) + L_{\text{bio}} \nabla \Sigma_{\text{bio}}(x) = K_{\text{bio}}(x - \mu) - L_{\text{bio}}(x - \mu).$$

Theorem 4 (Metriplectic scaffold law). *The finite scaffold law conserves total scaffold mass and produces nonnegative scaffold entropy:*

$$\frac{d}{dt} \sum_s x_s = 0, \quad \frac{d\Sigma_{\text{bio}}}{dt} = (x - \mu)^\top L_{\text{bio}}(x - \mu) \geq 0.$$

Proof. Write $y = x - \mu$. Since K_{bio} and L_{bio} have zero row sums, $\mathbf{1}^\top x = 0$. Since K_{bio} is skew-symmetric, $y^\top K_{\text{bio}} y = 0$. Since $\nabla \Sigma_{\text{bio}} = -y$,

$$\frac{d\Sigma_{\text{bio}}}{dt} = (-y)^\top (K_{\text{bio}} y - L_{\text{bio}} y) = y^\top L_{\text{bio}} y.$$

The graph Laplacian is positive semidefinite, so the entropy production term is nonnegative. $\square$

6.4 E_8 parity-sign geometry

Lemma 1 (Cardinality requires geometry). *A bare 240-element set carries cardinality. The E_8 root geometry requires an eight-dimensional inner product, squared norm two roots, prescribed inner-product multiplicities, and reflection closure.*

Proof. Cardinality is invariant under every bijection of a 240-element set. Inner products and reflections require additional structure. Therefore the E_8 geometry enters exactly when that structure is supplied. $\square$

Definition 5 (Parity-sign lift). Define

$$\begin{aligned} \mathcal{R}_{112} &= \{\pm e_i \pm e_j : 1 \leq i < j \leq 8\}, \\ \mathcal{R}_{128} &= \left\{ \frac{1}{2}(\epsilon_1, \dots, \epsilon_8) : \epsilon_i \in \{\pm 1\}, \prod_{i=1}^8 \epsilon_i = 1 \right\}, \end{aligned}$$

and

$$\mathcal{R}_{E_8} = \mathcal{R}_{112} \cup \mathcal{R}_{128}.$$

Theorem 5 (Full E_8 root geometry under parity-sign lift). *The set $\mathcal{R}_{E_8}$ has 240 vectors. Every vector has squared norm two. Every pairwise inner product lies in $\{-2, -1, 0, 1, 2\}$. Reflection through any root preserves $\mathcal{R}_{E_8}$. Hence the parity-sign lift realizes the full E_8 root geometry.*

Proof. The first family has $\binom{8}{2} \cdot 4 = 112$ vectors. The even half-sign family has $2^7 = 128$ vectors. A first-family root has squared norm 2, and a half-sign root has squared norm $8/4 = 2$. Coordinate overlap gives inner products in $\{-2, -1, 0, 1, 2\}$. For $\alpha \in \mathcal{R}_{E_8}$, the reflection

$$s_\alpha(\beta) = \beta - (\beta \cdot \alpha)\alpha$$

uses $\alpha \cdot \alpha = 2$. The companion SymPy audit enumerates all 240^2 pairs and all 240^2 reflected pairs, verifying membership closure exactly. $\square$

6.5 Projector residual laws

Let $r = (uv)^{-1}$, $\lambda_A = 2\pi A/3$, and $\delta = \pi r$. The four demonstration projectors are

$$\begin{aligned} \Pi_{\text{green}}(\widehat{\Xi}) &= \frac{1}{\sqrt{1+r^2}}(\cos \theta, \sin \theta, r \cos(\theta + \frac{\pi}{2}), r \sin(\theta + \frac{\pi}{2})), \\ \Pi_{\text{blue}}(\widehat{\Xi}) &= \begin{pmatrix} \cos \delta & -e^{-i(\theta+\lambda_A)} \sin \delta \\ e^{i(\theta+\lambda_A)} \sin \delta & \cos \delta \end{pmatrix}, \\ \Pi_{\text{red}}(\widehat{\Xi}) &= \tau_A = \frac{\theta + 2\pi A/3}{2\pi} + \frac{i}{uv}, \\ \Pi_{\text{yellow}}(\widehat{\Xi}) &= \left(\psi_A, Q_{\mu\nu} = g_{\mu\nu} - \frac{i}{2}\Omega_{\mu\nu}, \mathcal{S}_{240} \right), \quad \psi_A = \begin{pmatrix} \cos \delta \\ e^{i(\theta+\lambda_A)} \sin \delta \end{pmatrix}. \end{aligned}$$

For a branch law G_D , define

$$\mathcal{R}_D(X) = \|\Pi_D(\mathcal{E}X) - G_D(\Pi_D X)\|.$$

Theorem 6 (Quotient residual criterion). *A projected branch is an exact descendant of the lifted parent law on a regime when $\mathcal{R}_D(X) = 0$ throughout that regime. It is an effective descendant at scale ε_D when $\mathcal{R}_D(X) \leq C_D \varepsilon_D$ throughout the declared window.*

Proof. The residual is the norm of the commutator between lifted evolution followed by projection and projected evolution followed by the branch law. Exact descent is equality of those two routes. Effective descent is equality up to the declared scale of branch truncation, smoothing, or measurement. $\square$

6.6 Red contour equivalence

Theorem 7 (Red contour equivalence). *Let γ_1 and γ_2 be red-branch contours obtained from the same lifted operator word. Their red branch values are equal exactly when the difference of pulled-back completion one-forms is exact on the lifted word cylinder:*

$$\int_{\gamma_1} \omega_{\text{red}} - \int_{\gamma_2} \omega_{\text{red}} = 0 \iff [\omega_{\text{red}}^{(1)} - \omega_{\text{red}}^{(2)}] = 0 \in H_{\text{dR}}^1(\mathcal{C}_{\text{word}}).$$

The direct stereographic contour and the mock-modular completion contour coincide precisely in the zero-cocycle sector.

Proof. For closed one-form differences on a cylinder, equality of contour integrals over homologous representatives is equivalent to vanishing de Rham cohomology class. Assigning γ_1 to the stereographic correction contour and γ_2 to the red completion contour gives the zero-cocycle condition. $\square$

6.7 Negative-control theorem surface

Proposition 3 (Mandatory wrong-form failures). *The following stripped forms are rejected by exact witnesses:*

NC1. *Product collapse $q = (u, v) \mapsto uv$ identifies $(1, 6)$ and $(2, 3)$ although $\mathbf{B}(1, 6) = (6, 7)$ and $\mathbf{B}(2, 3) = (3, 5)$.*

NC2. *Sheet erasure removes κ , making visible $\mathbf{Q}^4$ recurrence indistinguishable from lifted return.*

NC3. *Residual relocation moves the unique 72-sector away from the phase-bearing successor.*

NC4. *Naming reversal violates $\mathbf{Q}$ - σ equivariance.*

NC5. *Cardinality alone supplies 240 without inner products or reflection closure.*

NC6. *Scalar red collapse to (π_m, B_m) forces the precision condition $b^{n+w-1}B_m < 1/2$ for radix-block certification.*

Proof. NC1 follows from the displayed unequal children. NC2 follows because θ is periodic while κ records lifted sheet number. NC3 follows from $N_{b_j}^{\text{cent}} = 40 + 16\omega(b_j)$. NC4 follows from the cyclic naming theorem. NC5 follows from the cardinality lemma. NC6 follows because a scalar interval for π has diameter $2B_m$, which becomes $2b^{n+w-1}B_m$ after multiplication by b^{n+w-1} ; certification requires that interval to fit inside one radix unit. $\square$

7 Native π Spigot Closure Surface

The same lifted object also supports an operational arithmetic readout of the primitive half-turn constant. The spigot branch is separated into three executable modes: native streaming prefix emission, certified bank-indexed random access after native construction, and no-bank local decoding.

Definition 6 (Native streaming spigot). A native streaming spigot is an executable readout that emits radix digits from the Phase Calculus carried state without importing an external oracle value of π . Its certificate is a pair

$$(D, D_{\text{safe}}), \quad D_{\text{safe}} \geq D,$$

where D is the emitted prefix length and D_{safe} is the certified lower bound on stable digits.

Definition 7 (Certified bank-indexed random access). Given a certified native prefix bank of length D and safe bound $D_{\text{safe}} \geq D$, a bank-indexed random-access query

$$(n, \ell) \mapsto d_n \dots d_{n+\ell-1}$$

is certified whenever $1 \leq n$ and $n + \ell - 1 \leq D$.

Theorem 8 (Certified native bank access). *If a native prefix bank has length D and certificate $D_{\text{safe}} \geq D$, then every indexed block query with $1 \leq n$ and $n + \ell - 1 \leq D$ returns a certified radix block.*

Proof. The emitted prefix is certified stable through at least D_{safe} digits. Since $D_{\text{safe}} \geq D$, every bank digit lies inside the certified stable prefix. The indexed query returns a substring of that certified prefix, so the returned block is certified. $\square$

Table 4: Native π spigot streaming certificates from the locked v8 artifact.

Digits	Mean seconds	Mean digits/s	Safe lower bound	Iterations
520	0.00002852	18,230,645.72	661	8
1,000	0.00006579	15,199,000.51	1,328	9
10,000	0.00254813	3,924,440.39	10,666	12
100,000	0.08884997	1,125,492.76	170,763	16
1,000,000	1.65086778	605,742.03	1,366,163	19

Proposition 4 (Locked v8 operational certificate). *The v8 native spigot artifact emits a certified one-million-decimal-digit native bank in mean wall time 1.65086778 seconds with certified safe lower bound 1,366,163 digits. The same artifact emits the 100,000-digit prefix in 0.08884997 seconds, corresponding to 1,125,492.76 digits/s on the recorded harness.*

The v8 bank-indexed random-access benchmark then performs certified indexed block queries over the native bank. The recorded query classes are 16, 64, 256, and 1024 digit blocks; all sample blocks lie inside the certified safe lower bound.

Table 5: Bank-indexed random access after native bank construction.

Block length	Queries	Mean seconds/query	Mean digits/s
16	2000	2.3020815×10^{-6}	6.95×10^6
64	2000	3.4344050×10^{-7}	1.86×10^8
256	2000	3.5544650×10^{-7}	7.20×10^8
1024	2000	5.3941900×10^{-7}	1.90×10^9

7.1 No-bank local decoding and scalar-collapse obstruction

The no-bank local decoder is a different mode. The v9 artifact builds a native warm-up state, orients the query, applies local block projection, and certifies requested decimal and hexadecimal probes at positions

$$1, 4895, 500000, 999937, 1000000.$$

All requested probes are certified in both decimal and hexadecimal, and the artifact marks the candidate path as no-prefix-bank.

Theorem 9 (Scalar-collapse certification bound). *Let a radix- b block decoder at position n and block length ℓ use only a scalar interval*

$$\pi \in [\pi_m - B_m, \pi_m + B_m].$$

If the decoder is to certify a block through this scalar interval alone, then it must satisfy

$$2b^{n-1}B_m < b^{-\ell},$$

equivalently

$$B_m < \frac{1}{2}b^{-(n+\ell-1)}.$$

If $B_m = 10^{-D_m}$, then the necessary decimal warm-up obeys

$$D_m > (n + \ell - 1) \log_{10} b + \log_{10} 2.$$

Proof. Multiplication by b^{n-1} scales the uncertainty interval for the digit block. If the scaled interval has diameter at least one radix block cell of size $b^{-\ell}$, two compatible values of π can occupy different blocks. Therefore certification by scalar interval alone requires

$$2b^{n-1}B_m < b^{-\ell}.$$

Solving gives $B_m < \frac{1}{2}b^{-(n+\ell-1)}$. Substituting $B_m = 10^{-D_m}$ and taking $\log_{10}$ gives the displayed lower bound. $\square$

Proposition 5 (v10/v11 structural closure). *The v10 adaptive live-decoder sweep realizes the scalar-collapse law. For decimal 32-digit probes it records*

$$D_m = n + 83$$

for $n = 1, 10, 100, 1000, 10000, 100000, 1000000$. With a fixed six-thousand-digit warm-up, positions 1 and 4,895 certify, while positions 500,000, 999,937, and 1,000,000 do not. The v11 theorem closes the general scalar-collapse conclusion for the current universal-collapse plus AGM family:

$$D_m = \Omega(n).$$

Corollary 1 (Branch-retaining decoder target). *The scalar-collapse obstruction applies only after reducing the lifted state to (π_m, B_m) . A native from-zero random-access decoder must retain branch-local structure from $(A, q, \theta, \kappa, c)$ with $q = (u, v)$, or from an exactly equivalent transported residue object. The next decoder law is therefore a branch-retaining map*

$$\mathcal{A}_b(n; \hat{\Xi}) \longrightarrow \mathcal{R}_{b,\ell}(\mathcal{A}_b(n; \hat{\Xi})) = (d_n, \dots, d_{n+\ell-1})_b,$$

not another scalar interval readout.

8 Unification Theorem

Theorem 10 (Unification theorem (demonstration form, v3)). *Let*

$$\hat{\Xi} = (A, q, \theta, \kappa, c), \quad q = (u, v), \quad c = \left[\theta - \frac{\pi}{uv}, \theta + \frac{\pi}{uv} \right],$$

be the exact lifted state on the balanced-window family, and let $\mathcal{E}$ be the single parent evolution generated by the Phase Calculus operator spine and its selector-closed executable macro-calculus. Then, on the declared regimes closed by the merged formalisation, the four-filter package, the native biological supplement, and the v3 extension closure:

- (i) *the green, blue, red, and yellow readouts are covariant projections of the same $\hat{\Xi}$ under the same operator grammar;*
- (ii) *the yellow quotient readout reproduces the CF01 quantum-geometric-tensor split into $g_{\mu\nu}$ and $\Omega_{\mu\nu}$ on its declared local chart;*
- (iii) *the later physics branches of the current CF stack descend from the same parent evolution by quotient projection and derivation-chain commutation on their declared regimes;*
- (iv) *the yellow biological readout derives, from the quarter alphabet generated by $\mathbf{Q}$, the triplet selector block, and the normalized edge residual $24E_{\text{edge}} = 1$, the scaffold*

$$240 = (128, 112) \rightarrow ((72, 56), (56, 56));$$
- (v) *inside the declared admissible yellow class, the biological weight projector and cyclic RNA naming are unique;*
- (vi) *the scaffold supports a finite metriplectic biochemical dynamics law with conserved total mass and nondecreasing scaffold entropy; and*
- (vii) *the full E_8 root geometry is realized by the parity-sign lift of the (112,128) scaffold split, while count-only E_8 determination is explicitly rejected.*

Therefore the current demonstration supports one precise native-source statement: the same singular lifted object $\hat{\Xi}$ is the common parent object for the present Phase Calculus hierarchy, for the physics branches descended from it, for the discrete biological scaffold derived on the yellow quotient, and for the parity-sign geometric lift that realizes the E_8 root system.

Proof sketch. The operator-core, selector-closed executable macro-calculus, arithmetic-completion bridge, and exact quotient criterion are supplied by the merged formalisation. The four-filter package evaluates four distinct projection families on the same retained state variables and shows exact-zero primary covariance errors, machine-precision recovery errors, and failure of stripped-state controls on the readouts that still require full lifted-state retention. The yellow filter is constructed so that its continuous local section is exactly

the QGT section used in the CF01 bridge. The biological scaffold follows from the yellow quotient itself: the quarter alphabet gives four states, the macro block gives $4^3 = 64$ triplet words, the normalized edge residual contributes one extra discrete channel, and the host labels index the three nontrivial $2 + 2$ partitions of the four-cycle.

The v3 extension closure adds the missing theorem surfaces. Admissible yellow uniqueness follows because the residual-localization and minimal positive integer constraints force $\omega(b_j) = 1 + \delta_{j1}$. Cyclic naming uniqueness follows from Q-equivariance and the seed $b_0 = u$. The finite biochemical dynamics law is a graph-metriplectic law on the 240-site scaffold; mass conservation follows from zero row/column sums and entropy production follows from positive semidefiniteness of the scaffold Laplacian. The E_8 geometry statement is not count-only: the count-only claim is false, and the full geometry is supplied by the parity-sign lift $\mathcal{R}_{112} \cup \mathcal{R}_{128} \subset \mathbb{R}^8$. These steps are attacked in the Lean, SymPy, and Jupyter companions. The theorem remains regime-qualified: it asserts native-source unification for the current closed stack and declared projector family, not uniqueness after erasing the admissibility contracts. $\square$

The theorem should be read together with its regime statement. The merged formalisation already allows exact or effective descent depending on whether the branch law is algebraic or windowed [1]. The four-filter package then adds the biological yellow layer and tests it on the exact same retained lifted state. The demonstration therefore closes native-source unification at the level appropriate to the present evidence: common parent object, common parent evolution, faithful multi-filter readout, and exact internally derived biological decomposition.

9 Implications and Completion Status

The immediate implication is architectural. Phase Calculus extensions are no longer added by inventing new primitive objects branch by branch. The primitive state is fixed, and every branch must state its projector, residual law, regime window, and negative control.

The completion extensions above change the publication surface in four ways. First, the biological projector is unique inside the declared admissible yellow class. Second, the cyclic RNA naming is unique under Q-equivariance and the seed convention. Third, the biological scaffold is no longer static only: it carries a finite metriplectic scaffold dynamics with exact mass conservation and nonnegative entropy production. Fourth, the E_8 statement is no longer count-only. Count-only determination is killed; the full root geometry is supplied by the parity-sign lift and attacked computationally. Fifth, the native π spigot surface is locked as an operational result with three separated modes: streaming prefix emission, bank-indexed certified random access, and no-bank local decoding with scalar-collapse obstruction isolated.

The remaining publication discipline is not a new open-theory burden. It is execution discipline: every future branch extension must preserve the same projector ledger and must carry a failure witness when the lifted ontology is stripped.

10 Closure Certificate

10.1 Assumption ledger

This paper uses seven assumption classes.

1. **Imported formal closure.** The primitive-source layer, the lifted object, the operator-core, the selector-closed executable macro-calculus, the exact quotient theorem, the continuous shadow theorem, and the derivation-chain exactness theorem are imported from the merged formalisation and the underlying *CF* papers [1, 4, 5].
2. **Imported computational closure.** The four-filter gate values, recovery errors, negative controls, and QGT residuals are imported from the executed four-filter package [20].
3. **Native biological derivation.** The quarter alphabet, the triplet selector block, the normalized residual bonus $24E_{\text{edge}} = 1$, the 240 total, the (56, 72, 56, 56) centered counts, and the three host-indexed (128, 112) bifurcations are derived inside the present paper and attacked in the Lean, SymPy, and Jupyter companions.

4. **Extension closure.** The admissible yellow projector uniqueness theorem, cyclic naming theorem, finite biochemical scaffold dynamics law, parity-sign E_8 lift, explicit projector ledger, contour-equivalence criterion, and negative controls are added in v3 and attacked in the new companion artifacts.
5. **External comparison benchmark.** The Halitsky–Klitzing–Moxness preprint is used only as an external comparison target for the codon:anticodon / E_8 count pattern [21].
6. **Demonstration-level synthesis.** The present paper contributes the compressed exposition, the cross-linking of the imported results, and the regime-qualified central unification statement.

10.2 Limits

This paper does *not* claim:

- (i) a re-proof of the entire formal stack,
- (ii) uniqueness of yellow projectors after removing the admissibility contract Y1–Y4,
- (iii) identity of direct stereographic correction with the mock-modular Eichler shadow outside the zero-cocycle case,
- (iv) empirical biochemical adequacy of the finite scaffold dynamics before dataset gates are run,
- (v) the full E_8 root geometry from count data alone.

It also keeps the same discipline as the Shadow Calculus supplement: completion/shadow data remain downstream of the primitive operator core and may not be promoted to source law by rhetoric alone [4].

10.3 Red-team checklist

A reviewer can now challenge the stack in five targeted places.

1. **Lifted-state sufficiency.** Show that the retained state $(A, q, \theta, \kappa, c)$ is not minimal or not sufficient.
2. **Selector closure.** Find a reachable executable balanced-window case not governed by the state-only selector.
3. **Quotient descent.** Produce a declared branch whose projector does not commute with the parent evolution even on its own stated regime.
4. **Four-filter covariance.** Reproduce the package and show a nonzero primary covariance failure, or demonstrate that one of the negative controls does *not* fail when stripped.
5. **Biological derivation.** Show that the native scaffold does *not* follow from the quarter alphabet, the triplet selector block, and the normalized residual bonus $24E_{\text{edge}} = 1$, or show that the distinguished doubled sector is not canonically fixed by the declared yellow chart.
6. **Extension closure.** Find an admissible yellow projector violating the uniqueness theorem, an admissible cyclic naming different from u, c, a, g , a scaffold-dynamics mass or entropy violation, or an E_8 root outside the parity-sign reflection closure.
7. **Native π spigot.** Reproduce the v8–v11 artifacts and show failure of the certified native bank, bank-indexed access bounds, no-bank probe certificates, or scalar-collapse obstruction theorem.

The point of the present paper is that the targets are now specific enough to attack.

A Benchmark tables

Table 6: Key four-filter quantitative results used in the paper

Quantity	Value imported from the executed four-filter package
Canonical anchor	$(u, v) = (55, 89)$, $uv = 4895$, $\delta_* = \pi/4895$, $\text{Im}(\tau_*) = 1/4895$
Primary covariance gates	Max green, blue, red, yellow error = 0 on short exhaustive, deep deterministic, and random-walk tests
Recovery gate	48 samples; largest reported δ -recovery residual $\approx 1.06 \times 10^{-100}$ (green/blue) and 6.98×10^{-105} (yellow)
Negative control 1	Product-only collapse: $(1, 6)$ and $(2, 3)$ share $uv = 6$ but have different B -images $(6, 7)$ and $(3, 5)$
Negative control 2	Sheet-loss witness: Q^4 visible recurrence is insufficient when κ is removed
QGT bridge	15 corridor rows; maximal closed-form residual $\approx 1.49 \times 10^{-101}$
Biological scaffold	Native identities checked in the release supplement: $4^3 = 64$, $24E_{\text{edge}} = 1$, weights $(1, 2, 1, 1)$, total 240, centered counts $(56, 72, 56, 56)$, and host partitions $(128, 112)$
Native π spigot	v8 emits 1,000,000 native decimal digits in 1.65086778 s with safe lower bound 1,366,163; bank-indexed access is certified inside the native bank; v10/v11 prove scalar-collapse warm-up obstruction
Extension closure	Release attacks: admissible projector uniqueness, cyclic naming uniqueness, finite scaffold dynamics, parity-sign E_8 root closure, contour-equivalence criterion, and six negative controls

B Minimal filter pseudocode

For orientation only, the four filters can be written schematically as

$$\Pi_{\text{green}}(\widehat{\Xi}) \rightarrow (S^3, \text{stereographic}), \quad \Pi_{\text{blue}}(\widehat{\Xi}) \rightarrow (SU(2), S^2),$$

$$\Pi_{\text{red}}(\widehat{\Xi}) \rightarrow \tau \text{ and shadow-attached branch data}, \quad \Pi_{\text{yellow}}(\widehat{\Xi}) \rightarrow (E(t), Q_{\mu\nu}, g, \Omega, \text{native 240-slot scaffold}).$$

The demonstration theorem requires faithful covariance, not numerical identity of all filters: each filter remains a readout of the same retained lifted state under the same operator grammar.

C Native biological derivation table

Table 7: Native biological scaffold identities derived in the release supplement

Identity	Exact value
Quarter alphabet size	4
Triplet selector block size	$4^3 = 64$ words
Normalized edge bonus	$24E_{\text{edge}} = 1$
Native weight budget	$(u, c, a, g) = (1, 2, 1, 1)$
Occurrences per base across the triplet block	48
Total scaffold size	$48(1 + 2 + 1 + 1) = 240$
Base-centered scaffold	$(u, c, a, g) = (56, 72, 56, 56)$
Host-indexed bifurcations	$(m, k) = (128, 112)$, $(y, r) = (128, 112)$, $(s, w) = (128, 112)$

References

- [1] Justin K. Lietz. *Phase Calculus: Complete Formalisation of the Primitive Roll, the Exact Lifted Object, the Lifted-State Operator Core, the Selector-Closed Executable Macro-Calculus, the Universal Lifted-State Calculus, and Shadow Calculus*. Supplied merged manuscript, 2026.

-
- [2] Justin K. Lietz. *Phase Calculus: Complete Formalism — Lifted-State Operator Calculus of the Primitive Roll*. 2026.
 - [3] Justin K. Lietz. *Phase Calculus: Connection-Closure Proofs*. Note, 2026.
 - [4] Justin K. Lietz. *Shadow Calculus: Supplement to Phase Calculus — Completion, Residual, and Launched-Family Branch Structure*. 2026.
 - [5] Justin K. Lietz. *CF 19: The Full Lifted Object of the Primitive Roll and Exact Return to Lifted Origin*. 2026.
 - [6] Justin K. Lietz. *CF 000: Primitive Distinguishability and Orthogonal Articulation*. 2026.
 - [7] Justin K. Lietz. *CF 00: Induced Geometry and Emergent Dynamics in VDM*. 2026.
 - [8] Justin K. Lietz. *CF 01: Quantum Geometric Tensor to Metriplectic Brackets in VDM*. 2026.
 - [9] Justin K. Lietz. *CF 04: Telegraph–Fisher Causality*. 2026.
 - [10] Justin K. Lietz. *CF 06: Information Geometry Foundations*. 2026.
 - [11] Justin K. Lietz. *CF 07: Measurement Theory Foundations for VDM*. 2026.
 - [12] Justin K. Lietz. *CF 08: Spinor Emergence via Domain-Wall Fermions in VDM*. 2026.
 - [13] Justin K. Lietz. *CF 09: Gauge Field Emergence via Berry Connection in VDM*. 2026.
 - [14] Justin K. Lietz. *CF 12: Gravity, Black Holes, and Saturation Geometry in VDM*. 2026.
 - [15] Justin K. Lietz. *CF 14: Principle of Stationary Action as Temporal Effective Invariant*. 2026.
 - [16] Justin K. Lietz. *CF 15: Noether’s Theorem as Invariant Articulation Cost Under Symmetry*. 2026.
 - [17] Justin K. Lietz. *CF 16: Electroweak Fiber Splitting, Chiral Weak Coupling, Neutral Gauge Mixing, and Broken Gauge–Scalar Structure in VDM*. 2026.
 - [18] Justin K. Lietz. *CF 17: Non-Abelian Saturation, Flux-Tube Formation, Quantitative Confinement, String Breaking, and Deconfinement in VDM*. 2026.
 - [19] Justin K. Lietz. *CF 18: Scale Re-Articulation, Effective Action, Hidden-Return Corrections, Running Couplings, Renormalization, and Corrected Symmetry Structure in VDM*. 2026.
 - [20] Justin K. Lietz. *phase_four_filter_package_v1*. Executed validation package with note, notebook, summary, and gate outputs, 2026.
 - [21] David Halitsky, Richard Klitzing, and Greg Moxness. *The H-bond Acceptor Scaffold for Watson–Crick Codon:Anticodon Pairing and the 4_{21} Polytope Scaffold for E_8* . Preprint supplied with the current package, 2026.
 - [22] Justin K. Lietz. *Phase Native Streaming Spigot v8, Local Decoder v9, Live Decoder v10, and Structural Closure v11*. Executed artifact bundles, 2026.